Math 322: Commutative Algebra

Chapter 1: Euclidean Domains and Principal Ideal Domains

- **Ring** Abelian group $(R, +)$ equipped with a multiplication that satisfies left and right distributivity property and contains a multiplicative identity
- **Commutative Ring** If the multiplication is commutative
- **Zero Divisor** Let R be a commutative ring. An element $a \in R\backslash\{0\}$ is a non-zero divisor if there exists $b \in R\backslash\{0\}$ such that $ab = 0$.
- **Integral Domain** A commutative ring with no zero divisors, written ID.
- *Property of ID* Multiplicative Cancellation: If $0 \neq a \in R$ and $b, c \in R$ such that $ab = ac$ then $b = c$.
- **Ideal** A subgroup I of $(R, +)$ such that $ab, ba \in I$ for any $a \in R$ and $b \in I$.
- **Generating Set** A generating set for an ideal I is a subset S of I, such that the elements of S R-generate S.
- **Ideal of R generated by S** If $S = \{s_1, s_2, \cdots s_n\}$ is finite then $I = s_1R + s_2R + \cdots s_nR$ is the ideal of R generated by S.
- **Principal** An ideal I is principal if I can be generated by single element.
- Given two ideals I, J in a commutative ring R, we can construct new ideals:

$$I \cap J,$$

$$I + J,$$

$$IJ = \left\{ \sum_{\text{finite}} ab : a \in I, b \in J \right\}$$

- *Notation.* We write $a \equiv b \mod I$ for $a - b \in I$.
- **Principal Ideal Domain** An integral domain such that any ideal is principal.

- Main instances of PID's:
  1. Ring of Integers
  2. Polynomial rings in one variable over a field
- **Euclidean Function** Let R be an integral domain. A Euclidean Function on R is a function $v : R\backslash\{0\} \to \mathbb{N}_0$ sucht that the following hold:
  (i) $v(ab) \geq v(a)$ for all $a, b \in R\backslash\{0\}$
  (ii) for any $a \in R$ and any $b \in R\backslash\{0\}$, there exists $q, r \in R$ such that $a = qb + r$ and either $r = 0$ or $v(r) < v(b)$
- **Euclidean Domain** ID R such that there is a Euclidean Function on R.
- *Important ED:* Ring of Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ paired with the Euclidean Function

$$v : \mathbb{Z}[i]\backslash\{0\} \longrightarrow \mathbb{N}_0, v(z) = |z|^2 \quad \text{for all} \quad z \in \mathbb{Z}[i]$$

- ED $\implies$ PID
- *Important theorem relating a ring and its polynomial ring* Let R be a ring and $R[x]$ the polynomial ring over R. TFAE
  (i) $R[x]$ is a ED
  (ii) $R[x]$ is a PID
  (iii) $R[x]$ is a field
- *Non-example to above theorem* Consider the ring $\mathbb{Z}[x]$ of polynomials with integer coeffiecients. Recall that $\mathbb{Z}$ is not a field. Show $\mathbb{Z}[x]$ is not a ED by showing second axiom of ED fails using the functions $f = x^3 - 3x + 1$ and $g = 2x$.

  In the next three definitions, let R be a commutative unital ring, with multiplicative identity 1 and additive identity 0.
- **Inveritible** An element $a \in R$ is invertible if there exists $b \in R$ such that $ab = 1$.

  $R^\times$ is the set of all invertible elements of R and a^{-1} is the inverse of a.
- **Divides/Factor** If $a, b \in R$ are both non-zero, we say a divides b a is a factor of b if there exists $c \in R$ such that $b = ac$. Write $a|b$.

- **Associated** Two non-zero elements $a, b \in R$ are associated if only if $a|b$ **and** $b|a$. Write $a \sim b$.
- *Useful divisibility proposition* Let R be a commutative ring and $a, b \in R\backslash\{0\}$. Then
$$a|b \iff bR \subseteq aR.$$
When considering questions of divisibility, we will generally work with IDs only.
- *Remarks*
  - We can also refer to an invertible element as a **unit** in a ring.
  - $u \in R^\times \iff uR = R$
  - $R^\times$ forms a mulitplicative group
  - The relation 'a is associated to b in R' is an equivalence relation
- *Characterisation of invertible elements of a polynomial IDs* Let R be an ID. Then $R[x]^\times = R^\times$ (All the invertible elements in $R[x]$ are the constant polynomials.
- *Characterisation of invertible elements of a ED* Let R be a ED with Euclidean function v. The following hold
  (i) $v(1) = \{v(a) : a \in R\backslash\{0\}\}$

  (ii) $a \in R$ is invertible if and only if $v(a) = v(1)$.